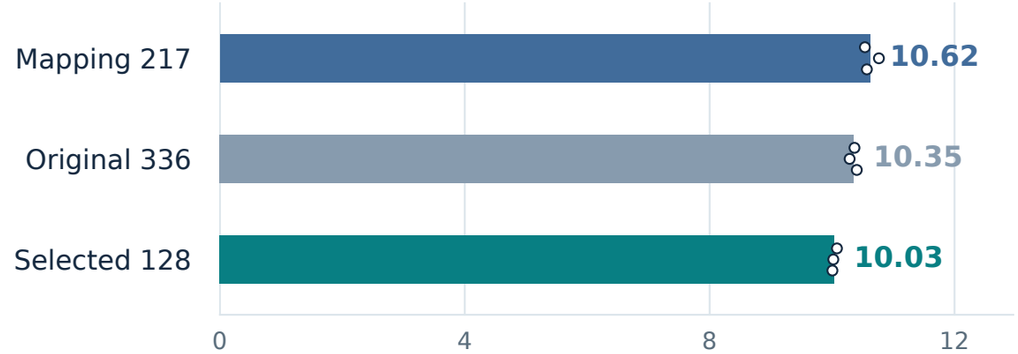
Development gains did not transfer to the fresh check

Matched feature views · lower MAE is better · cohorts evaluated separately

A Development selection stability

Mean group-macro MAE across three seeds · LUT4

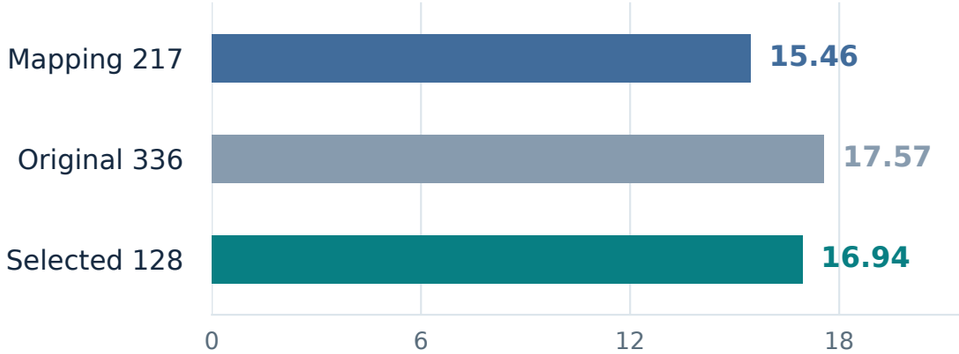


128 vs 217: -5.53% MAE; lower on all three seeds.

1,665 rows · 522 groups · seed dots · same-fold Jaccard 0.845

B Prospective synthetic template check

Mean absolute error on 40 newly measured circuits · LUT4



128 vs 217: +9.60% MAE; fresh success rule failed.

Five equal-size templates · MAPE: 14.09% → 15.56% (217 → 128)

C Hardware budget decisions: baseline 217 → selected 128

Development: sums over three descriptive seed repeats

Cap (LUT4)	False-safe: 217 → 128	False rejection: 217 → 128
32	**65 → 60**	**289 → 278**
64	**45 → 56**	**115 → 121**
128	**31 → 38**	**19 → 21**

Worst underprediction: 193.59 → 194.72 LUT4

Development strict eligibility: FAIL

Fresh: 40 frozen predictions matched to new labels

Cap (LUT4)	False-safe: 217 → 128	False rejection: 217 → 128
32	**0 → 0**	**7 → 8**
64	**6 → 7**	**0 → 0**
128	**0 → 1**	**0 → 0**

Worst underprediction: 52.15 → 56.05 LUT4

Fresh predeclared success: FAIL

Selected 128 remains experimental; keep 217 as the deployed default. No calibrated budget guarantee.

False-safe: prediction ≤ cap, measured area > cap. Seed repeats reuse the same consumed cohort; they are not independent samples.

Fresh scope: five synthetic circuit templates; no independent application-transfer or full-core claim. Sources: results/selection-stability and results/selection-validation.